

Table 1. Comparison of HALO with prior synergy prediction frameworks

Method	Model / architecture	Feature type	Output	Dataset scale	Strain	Evaluation scheme	Reported metrics
CoSynE [16]	Random Forest	Substructure fingerprints	3-class (syn/add/antag)	153 pairs, 18 cmpds, 1 strain	<i>E.coli</i> MG1655	K-fold / leave-drug-out / leave-pair-out	$Acc = 0.81$, $AU - ROC = 0.77$ (leave-pair-out)
INDIGO [17]	Random forest	Chemogenomic profiles	Continuous Loewe α	105–171 pairs, 15–19 cmpds (in train–test), 1 strain	<i>E.coli</i> MG1655	Leave-drug-out / leave-pair-out	$R = 0.52$ ($P = 10^{-6}$) for 66-pair test
Lv et al. [18]	Graph regularization + network propagation	Protein-protein interaction network + drug targets	Binary	91 pairs, 14 cmpds, 1 strain	<i>E.coli</i> MG1655	Not confirmed	$Acc = 0.90$, $Precision \approx 0.875$
HALO	LightGBM	Multi-level CC bioactivity embeddings	Binary	2,449 interactions, 97 cmpds, 10 strains	Nested pair-held-out CV	$ROC - AUC = 0.78$, $Accuracy = 0.70$ (Sec. 2.2)	